

Round-23: the Catalogue Generator in Lean

“Every unfolding of the catalogue” becomes a theorem about a syntactic object

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Abstract

Round-22 lifted the S-condition to catalogue data (**SCat**) but still required, per certificate, that it be *wellformed* and *faithful*. Round-23 discharges both by construction. `formal/Round19Transport.lean` (4,056 lines, still **zero sorry**) now defines a **Catalog** — templates, N2 attachment rules (each child slot maps to a parent member port, never core), a root, and template-indexed steering — and a builder **buildCert** that turns a *plan* (a list of rule/parent choices) into a certificate. The round-23 theorems prove that for any structurally sound catalogue (**CatOk**) and any valid plan (**PlanOk**), the built certificate is **Wellformed** (`build_wellformed`) and **Faithful** (`build_faithful`) — so `catalogue_soundness` derives the full per-unfolding **SCond** for *every* plan from the single catalogue check **SCat**. Faithfulness — the thirteenth review’s parent-pattern agreement, the last per-certificate hypothesis — is proved by a parent-chain induction from each rule’s port-agreement condition. “Every unfolding of the catalogue” is thereby a theorem about a syntactic object rather than a hypothesis pair. A concrete catalogue witness (**certK**) drives the entire rounds 19–23 pipeline to a closed frame with no per-certificate proof. Status: *strongly supported, not certified*.

1 What was proved

- **Catalog**, **Attach** (child template, expected parent template, N2 slot map), and **buildCert**: a plan folds attachment rules onto the root, each child inheriting its parent’s member occurrences through the slot map. Indexing lemmas (`buildSteps_length`, `buildCert_step_zero`, `buildCert_step_succ`) pin what each built step is.
- **CatOk** / **AttachOk** / **PlanOk**: structural soundness of the catalogue, of each rule (valid templates, injective slot map into parent member ports, and *port agreement* — old child ports and their parent images carry equal net values), and of a plan (real rules, earlier parents, matching parent templates).
- `build_wellformed`: every planned certificate satisfies all seven **Wellformed** clauses — targets pre-exist, are born and in range, one per slot, injective; templates valid; nets converse-coherent; steering reads only rows. Proved by induction over the step prefix; slot injectivity descends from the rule’s `map_nodup` through the port map.
- `build_faithful`: co-members of a built step carry the step template’s net entry. Proved by a *parent-chain induction*: each old child port maps to a parent member port (`built_port_parent`),

*Round-23 of the $\mathcal{ALCT}_{RCC5}$ decidability project. Substrate: the rounds 19–22 Lean artifact (transport, characterization, S-layer closure, catalogue conditions). Ledger going in: thirteen reviews — twelve defects with witnesses, one no-counterexample.

the parent pair value is the parent template’s entry by the inductive hypothesis, and the rule’s **agree** condition transports it to the child template — exactly the thirteenth review’s parent-pattern agreement, now structural.

- **catalogue_soundness** (ROUND-23 MAIN THEOREM) and **catalogue_frame_closed**: **CatOk** + **PlanOk** + **SCat** give **SCond**, hence closed frames, for every plan.
- Witness **certK**: a root template (two fresh ports) and one rule whose slot inherits a root fresh port. **certK_catok** / **certK_planok** / **certK_scat** drive **certK_scond** and a closed-frame **example** through **buildCert**, with no per-certificate wellformedness or faithfulness proof written.

2 What remains

With round-23 the soundness pipeline is complete at the relational level: from catalogue syntax to closed atomic frames, every step kernel-checked. The remaining soundness work is the logic layer (round-24: $\mathcal{ALCI}_{RCC5}$ syntax and semantics over frames, Hintikka types, one-step demand fulfilment, the truth lemma) and the limit passage (round-25: finite prefixes to the full presentation, absorbing the RCC5 patchwork property). The completeness direction (extraction from Theorem-A presentations) still contains the open mathematics F6 (width budgets) and W2’ (uniformization). Ledger: thirteen reviews — twelve defects, one no-counterexample; rounds 19–23 unreviewed. The decidability theorem for $\mathcal{ALCI}_{RCC5}$ remains **strongly supported, not certified**.